Ballistic full-shell bottleneck

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Abstract Spatial profile of pairing amplitudes across a bottleneck in a simplified full-shell nanowire, with emphasis in the case with different fluxoids on either side

Without self-consistency

We define the system as a central region with leads. We make the two radii equal, changing only the fluxoids n_L and n_R on either side for simplicity.

First step, a Quantica tight-binding model

```
Nϕ = 56
R = 70                  # nm
a0 = R * 2sin(π/Nϕ)     # nm
zmax = 15*a0            # nm
const ħo2m = 76.1996    # mev nm

circle = [R*SA[cos(2π*n/Nϕ), sin(2π*n/Nϕ), 0] for n in 0:(Nϕ-1)]
lat = lattice(sublat(circle), bravais=SA[0, 0, a0])
Δf(r, n) = cis(n * atan(r[2], r[1]))
Δf(r, nL, nR) = Δf(r, ifelse(r[3] > 0.5*a0, nR, nL))
modelΔ = @onsite((r; t=ħo2m/a0^2, μ=0, nL=0, nR=0, Δ = 0.2, Δz=Returns(Δ)) ->
    let Δ = Δz(r[3]) * Δf(r, nL, nR)
        SA[4t-μ Δ; Δ' μ-4t]
    end)
modelt = @hopping((; t=ħo2m/a0^2) -> SA[-t 0; 0 t], range = a0)
model = modelt + modelΔ;
```

Now we carve out central region plus two leads

```
gL = translate(lat, SA[0, 0, -zmax]) |> hamiltonian(model, orbitals=2) |>
greenfunction(GS.Schur(boundary=0))
gR = translate(lat, SA[0, 0, zmax]) |> hamiltonian(model, orbitals=2) |>
greenfunction(GS.Schur(boundary=0))
```

```
g = lat |> supercell(region=r -> -zmax <= r[3] <= zmax) |> hamiltonian(model,
orbitals=2) |>
    @hopping!((t, r, dr; τ=1.0) -> t * τ; region = (r, dr) -> r[3] ≈ 0.5*a0) |>
    attach(gL[cells=-1], modelt, region=r -> r[3] ≈ -zmax) |>
    attach(gR[cells=1], modelt, region=r -> r[3] ≈ zmax) |>
    greenfunction;
```

```
┌ Warning: Renamed repeated sublattice :A to :B
└ @ Quantica ~/.julia/dev/Quantica/src/types.jl:62
┌ Warning: Renamed repeated sublattice :A to :B
└ @ Quantica ~/.julia/dev/Quantica/src/types.jl:62
```

We now want to plot the anomalous Green's function in space at zero frequency. To this end we compute a diagonal slice of $g(\omega = 0)$, and read out the corresponding site positions

```
function anomalous(func=abs; kw...)
    f = g[diagonal(kernel=SA[0 0; 1 0])](0.0; kw...)
    fd = func.(diag(f))
    zs = last.(sites(orbaxes(f)[1])) .- 0.5*a0
    fig = Figure(; plotkw[:figure]..., size = (600, 300))
    ax = Axis(fig[1, 1], xlabel=L"z/a_0", ylabel="$func[f(z,z)]")
    for nϕ in 1:Nϕ
        inds = nϕ:Nϕ:length(zs)
        lines!(ax, view(zs, inds), view(fd, inds))
        scatter!(ax, view(zs, inds), view(fd, inds))
    end
    m, M = 1.1 .* extrema(func, fd)
    ylims!(ax, min(0, m), max(0, M))
    xlims!(ax, -zmax + 1, zmax)
    return fig
end;
```

We plot the absolute value of the anomalous Green function with $n_L = 0, n_R = 0$. The result is uniform, as expected

```
anomalous(abs; nL=0, nR=0, μ = 0.05, Δ = 0.05)
```

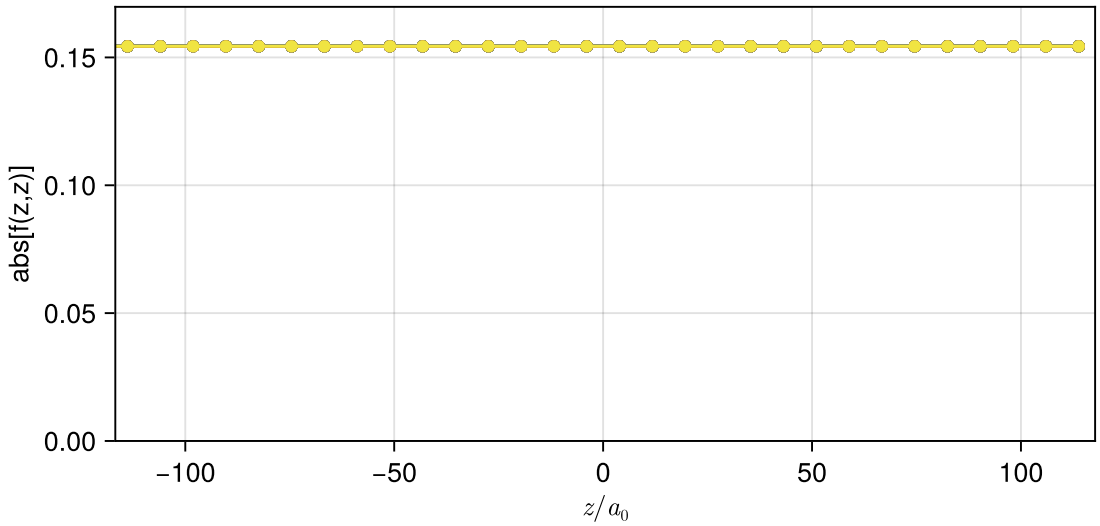


Figure 1: Anomalous Green function for $n_L = n_R = 0$

Now with $n_L = 1, n_R = 0$. The result is non-uniform now, and varies around the cylinder. The different curves correspond to different angle ϕ around the axis.

```
anomalous(abs; nL=1, nR=0, μ = 0.05, Δ = 0.05)
```

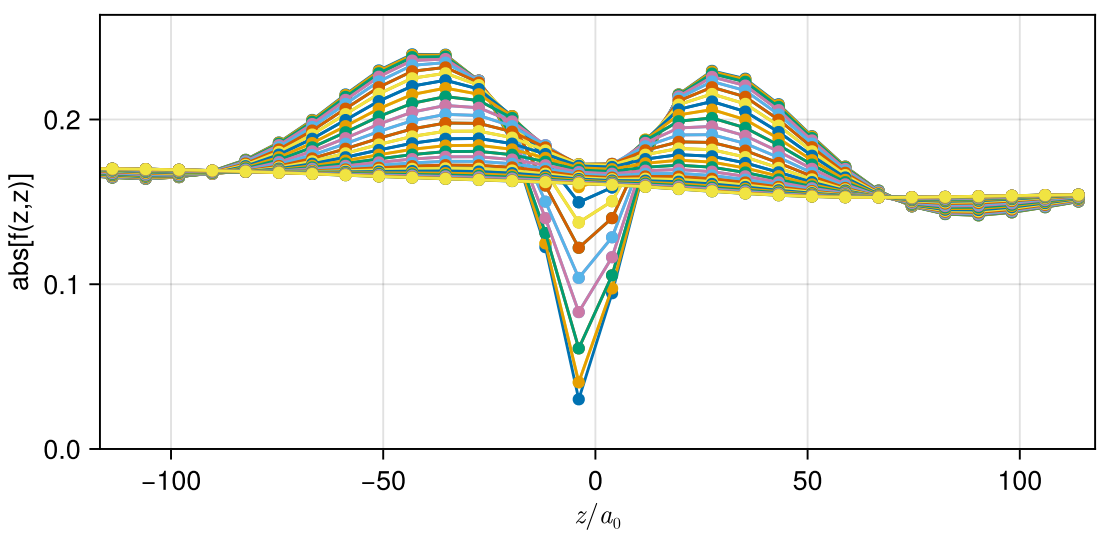


Figure 2: Absolute value of the anomalous Green function for $n_L = 1, n_R = 0$. The different colors correspond to different cylindrical angles ϕ.

```
anomalous(real; nL=1, nR=0, μ = 0.05, Δ = 0.05)
```

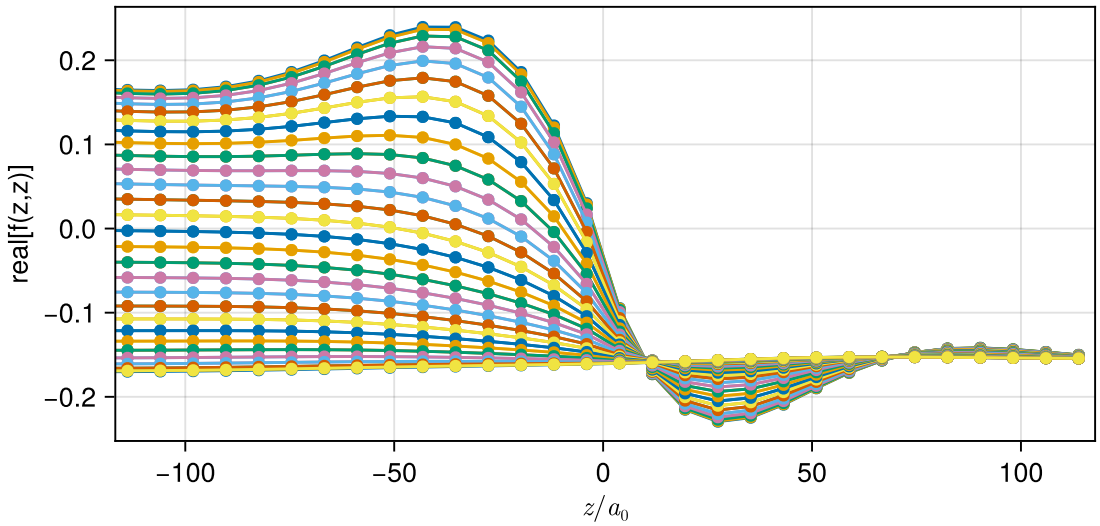
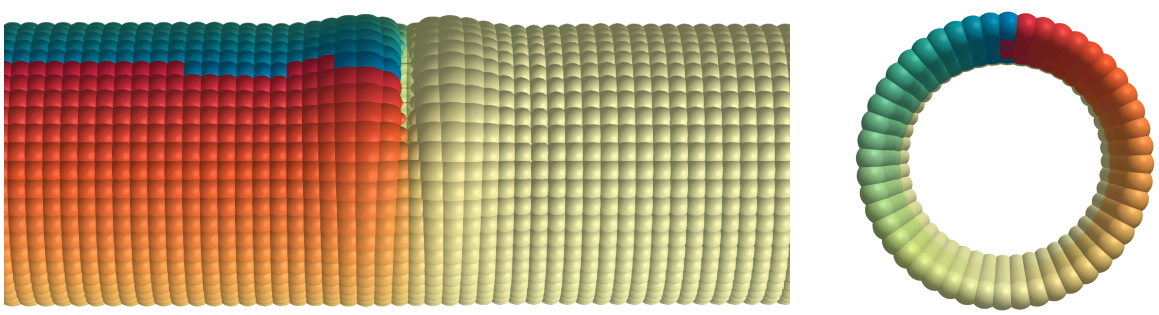


Figure 3: Real part of the anomalous Green function for $n_L = 1, n_R = 0$. Note that the solution is continuous and differentiable

Note that all ϕ cuts above are continuous and differentiable (as complex functions), but the solution is not ϕ independent. Notably, some ϕ's are forced to go to zero in absolute value at the junction $z = 0$. This is not suprising, since at some ϕ's there is a π jump in the superconducting phase at $z = 0$, whilst in others there is no jump. This reveals a crucial point: we cannot assume that $f(z, \phi) = |f(z)|e^{in(z)\phi}$, since the coefficient $|f(z)|$ also depends on ϕ.

Let's plot this in 3D.



(a) Tube side (b) Tube front

Figure 4: Anomalous Green function $f(r, r)$ for $n_L = 1, n_R = 0$. Colors correspond to the phase of f, and site radius to modulus $|f|$. Note that the change in fluxoid creates a topological hole on one side.

Note that the fluxoid frustration induces a hole, if $with zero pairing at a certain point on the side of the nanotube. The number of holes can be bigger, and corresponds to $|n_R -$

$n_L|$. The hole positions are only determined by the boundary conditions, and could move around, but would carry with them the phase string (blue/red boundary above). Will self-consistency change this? Probably not, as the hole is probably an unavoidable topological defect.